

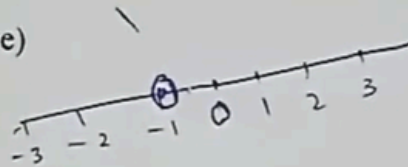
MCQs

Past Paper Midterm

PAPER NO.2

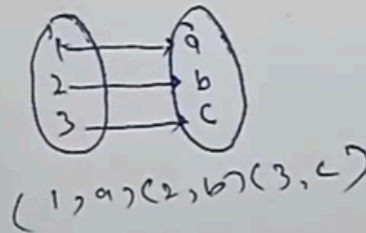
1. Every real number corresponds to _____ on the co-ordinate line.

- Infinite number of points
- Two points (one positive and one negative)
- A unique point???
- None of these



2. There is one-to-one correspondence between the set of points on co-ordinate line and _____

- Set of real numbers
- Set of integers
- Set of natural numbers
- Set of rational numbers

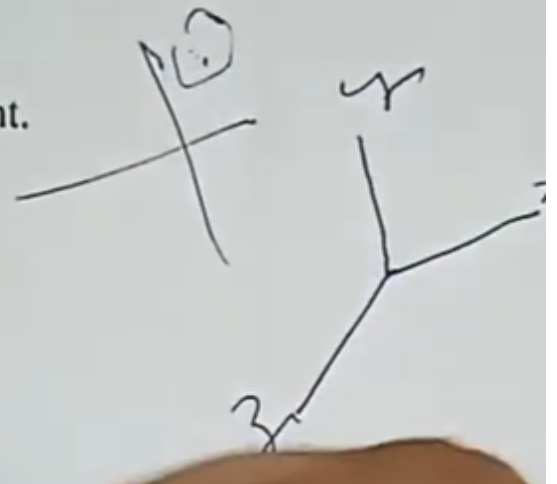


3. Which of the following is associated to each point of three dimensional spaces?

- A real number
- An ordered pair
- An ordered triple

4. All axes are positive in _____ octant.

- First
- Second
- Fourth
- Eighth



> Eighth

The spherical co-ordinates of a point are $(\sqrt{3}, \frac{\pi}{3}, \frac{\pi}{2})$. What are its cylindrical co-ordinates?

cylindrical spherical
 (r, θ, z) (ρ, θ, ϕ)

> $(\frac{\sqrt{3}}{2}, \frac{3}{2}, 0)$

> $(\sqrt{3} \cos \frac{\pi}{3}, \sqrt{3} \sin \frac{\pi}{3}, 0)$

> $(\sqrt{3} \sin \frac{\pi}{3}, \frac{\pi}{2}, \sqrt{3} \cos \frac{\pi}{3})$

> $(\sqrt{3}, \frac{\pi}{3}, 0)$

$$\begin{aligned} r &= \rho \sin \phi \\ r &= \sqrt{3} \sin \frac{\pi}{2} \\ &= \sqrt{3} (1) \\ r &= \sqrt{3} \end{aligned} \quad \left| \begin{aligned} \theta &= \theta \\ \theta &= \frac{\pi}{3} \end{aligned} \right| \quad \begin{aligned} z &= \rho \cos \phi \\ z &= \sqrt{3} \cos \frac{\pi}{2} \\ &= \sqrt{3} (0) \\ z &= 0 \end{aligned}$$

6. Suppose $f(x, y) = xy - 2y^2$ where $x = 3t + 1$ and $y = 2t$. Which one of the following is true?

> $\frac{df}{dt} = -4t + 2$

> $\frac{df}{dt} = -16t - 1$

> $\frac{df}{dt} = 18t + 2$

> $\frac{df}{dt} = -10t^2 + 8t + 1$

$$\begin{aligned} f(x, y) &= xy - 2y^2 \\ &= (3t+1)(2t) - 2(2t)^2 \\ &= 6t^2 + 2t - 8t^2 \\ &= -2t^2 + 2t \\ \frac{d}{dt}(-2t^2 + 2t) &= -2(2t) + 2(1) \\ &= -4t + 2 \end{aligned}$$

7. Let $w = f(x, y, z)$ and $x = g(r, s)$, $y = h(r, s)$, $z = k(r, s)$ then by chain rule $\frac{\partial w}{\partial r} =$

> $\frac{\partial w}{\partial r} = -10t^2 + 8t + 1$

7. Let $w = f(x, y, z)$ and $x = g(r, s)$, $y = h(r, s)$, $z = k(r, s)$ then by chain rule $\frac{\partial w}{\partial r} =$

> $\frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r}$

8. Magnitude of vector $\vec{a}$ is 2, magnitude of vector $\vec{b}$ is 3 and angle between them when placed tail to tail is 45 degrees. What is $\vec{a} \cdot \vec{b}$?

➤ 4.5

➤ 6.2

➤ 5.1

➤ 4.2

$$\begin{aligned} |\vec{a} \cdot \vec{b}| &= |\vec{a}| |\vec{b}| \cos \theta \\ &= (2)(3) \cos 45 \\ &= 6 (0.7071) \end{aligned}$$

9. Is the function $f(x, y)$ continuous at origin? If not, why?

$f(x, y) = \begin{cases} 0 & \text{if } x \geq 0 \text{ and } y \geq 0 \\ 1 & \text{otherwise} \end{cases}$

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➤ $f(x, y)$ is continuous at origin

➤ $f(0, 0)$ is not defined

➤ $f(0, 0)$ is defined but $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist

➤ $f(0, 0)$ is defined and $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ exist but these two numbers are not equal.

$$\begin{aligned} &= 4.24 \\ &\infty = \frac{1}{0} \\ &2 = \sqrt{-} \end{aligned}$$

10. Is the function $f(x, y)$ continuous at origin? If not, Why?

$$f(x, y) = \begin{cases} \frac{3x^2y}{x^2+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

➤ $f(x, y)$ is continuous at origin

➤ $f(0, 0)$ is not defined

➤ $f(0, 0)$ is defined but $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist

$$\frac{3x^2y}{x^2+y^2} = \frac{3x^2y}{0^2+0^2} = \frac{0}{0} = \infty$$

11. Let R be a closed region in two dimensional space. What does the double integral over R calculates?

➤ Area of R

➤ Radius of inscribed circle in R .

➤ Distance between two endpoints of R .

➤ None of these

12. Which of the following formula

12. Which of the following formula can be used to find the volume of a parallelepiped with adjacent edges formed by the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$?

➤ $|\vec{a} \times (\vec{b} \times \vec{c})|$

➤ $|\vec{a} \cdot (\vec{b} \cdot \vec{c})|$

➤ $|\vec{a} \cdot (\vec{b} \times \vec{c})|$

➤ $|\vec{a} \times (\vec{b} \cdot \vec{c})|$

$$(\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$\vec{a} \cdot (\vec{b} \times \vec{c})$$

13. Two surfaces are said to be orthogonal at a point of their intersection if their normals at that point are _____.

➤ Parallel

➤ Perpendicular

➤ In opposite direction

➤ Same direction

14. By Extreme Value Theorem, if a function $f(x, y)$ is continuous on a closed and bounded set R , then $f(x, y)$ has both _____ on R .

Absolute maximum and absolute minimum value

- Relative maximum and relative minimum value
- Absolute maximum and relative minimum value
- Relative maximum and absolute minimum value

15. Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy}$ and $f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let

$D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$ if

$D > 0$ and $f_{xx}(x_0, y_0) < 0$ then f has _____.

- Relative maximum at (x_0, y_0) $D > 0$ $f_{xx}(x_0, y_0) < 0$
- Relative minimum at (x_0, y_0) $D > 0$ $f_{xx}(x_0, y_0) > 0$
- Saddle point at (x_0, y_0) $D < 0$
- No conclusion can be drawn. $D = 0$

$D = 0$

16. Let the function $f(x, y)$ has continuous second-order partial derivatives $(f_{xx}, f_{yy}$ and $f_{xy})$ in some circle centered at a critical point (x_0, y_0) and let $D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$ if $D = 0$ then _____.

- f has relative maximum at (x_0, y_0)
- f has relative minimum at (x_0, y_0)
- f has saddle point at (x_0, y_0)
- No conclusion can be drawn.

➤ No conclusion can be drawn.

17. If $R = R_1 \cup R_2$, where R_1 and R_2 are non overlapping regions then

$$\iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA =$$

➤ $\iint_R f(x, y) dA$

➤ $\iint_{R_1} f(x, y) dA \cup \iint_{R_2} f(x, y) dA$

➤ $\iint_R f(x, y) dV$

➤ $\iint_R f(x, y) dV \cap \iint_{R_2} f(x, y) dA$

$$\begin{aligned} & \iint_{R_1} f(x, y) dA + \iint_{R_2} f(x, y) dA \\ &= \iint_R f(x, y) dA \end{aligned}$$

18. If $R = \{(x, y) / 0 \leq x \leq 2 \text{ and } 1 \leq y \leq 4\}$, then $\iint_R (6x^2 + 4xy^3) dA =$

~~$\int_1^4 \int_0^2 (6x^2 + 4xy^3) dy dx$~~

$\int_0^2 \int_1^4 (6x^2 + 4xy^3) dy dx$

➤ $\int_1^4 \int_0^2 (6x^2 + 4xy^3) dy dx$

➤ $\int_2^4 \int_0^1 (6x^2 + 4xy^3) dy dx$

$$\int_1^4 \int_0^2 (6x^2 + 4xy^3) dx dy$$

19. If $R = \{(x, y) / 2 \leq x \leq 4 \text{ and } 0 \leq y \leq 1\}$, then $\iint_R (4xe^{2y}) dA =$

$$\int_0^1 \int_2^4 (4xe^{2y}) dy dx$$

$$\int_2^4 \int_0^1 (4xe^{2y}) dx dy$$

$$\int_0^1 \int_2^4 (4xe^{2y}) dx dy$$

$$\int_2^4 \int_0^1 (4xe^{2y}) dy dx$$

20. If $R = \{(x, y) / 0 \leq x \leq 4 \text{ and } 0 \leq y \leq 9\}$, then $\iint_R (3x - 4x\sqrt{xy}) dA =$

$$\int_0^9 \int_0^4 (3x - 4x\sqrt{xy}) dy dx$$

$$\int_0^4 \int_0^9 (3x - 4x\sqrt{xy}) dx dy$$

$$\int_0^9 \int_0^4 (3x - 4x\sqrt{xy}) dx dy$$

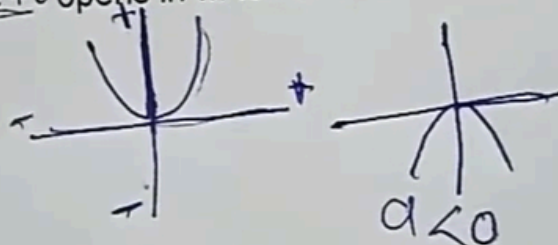
Question No: 1 (Marks: 1) - Please choose one
Which of the following number is associated to each point on a co-ordinate line?

- ▶ An integer
- ▶ A real number
- ▶ A rational number
- ▶ A natural number

Question No: 2 (Marks: 1) - Please choose one

If $a > 0$, then the parabola $y = ax^2 + bx + c$ opens in which of the following direction?

- ▶ Positive x - direction
- ▶ Negative x - direction
- ▶ Positive y - direction
- ▶ Negative y - direction



(Marks: 1) - Please choose one

Question No: 3 (Marks: 1) - Please choose one

Rectangular co-ordinate of a point is $(1, \sqrt{3}, -2)$. What is its spherical co-ordinate?

- ▶ $(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{2})$
- ▶ $(2\sqrt{2}, \frac{\pi}{2}, \frac{3\pi}{4})$
- ▶ $(2\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4})$
- ▶ $(\sqrt{2}, \frac{\pi}{3}, \frac{3\pi}{4})$

$(x, y, z) \rightarrow (\rho, \theta, \phi)$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\rho = \sqrt{1^2 + (\sqrt{3})^2 + (-2)^2} = \sqrt{8}$$

$$\boxed{\rho = 2\sqrt{2}}$$

$$\tan \theta = \frac{y}{x}$$

$$\theta = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right)$$

$$\theta = 60^\circ \times \frac{\pi}{180}$$

$$\boxed{\theta = \frac{\pi}{3}}$$

$$\cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$= \frac{-2}{2\sqrt{2}}$$

$$\cos \phi = -0.7071$$

$$\phi = \cos^{-1}(0.7071)$$

$$\phi = 135^\circ = \frac{3\pi}{4}$$

Question No: 4 (Marks: 1) - Please choose one

If a function is not defined at some point, then its limit: _____ exist at that point.

- ▶ Always
- ▶ Never

Question No: 4 (Marks: 1) - Please choose one

If a function is not defined at some point, then its limit: _____ exist at that point.

☐ Always

☒ Never

☐ May

Question No: 5 (Marks: 1) - Please choose one

Suppose $f(x, y) = x^3 e^{xy}$. Which one of the statements is correct?

☐ $\frac{\partial f}{\partial y} = 3x^3 e^{xy}$

☐ $\frac{\partial f}{\partial y} = x^3 e^{xy}$

☒ $\frac{\partial f}{\partial y} = x^4 e^{xy}$

☐ $\frac{\partial f}{\partial y} = x^3 y e^{xy}$

$$\frac{d}{dx} x^3 e^{xy} = x^3 \frac{d}{dx} (e^{xy}) = x^3 e^{xy} \cdot y = x^3 y e^{xy}$$

Question No: 6 (Marks: 1) - Please choose one

If $f(x, y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2} =$

Question No: 6 (Marks: 1) - Please choose one

If $f(x, y) = x^2 y - y^3 + \ln x$

then $\frac{\partial^2 f}{\partial x^2} =$

☐ $2xy + \frac{1}{x^2}$

☐ $2y + \frac{1}{x^2}$

☐ $2xy - \frac{1}{x^2}$

☒ $2y - \frac{1}{x^2}$

$$\frac{\partial f}{\partial x} = 2xy - 0 + \frac{1}{x}$$

$$\frac{\partial^2 f}{\partial x^2} = 2(1)y + \left(-\frac{1}{x^2}\right) = 2y - \frac{1}{x^2}$$

Question No: 7 (Marks: 1)

are not equal. $(x, y) \rightarrow (0, 0)$ exists but these two

Question No: 9 (Marks: 1) - Please choose one

What is the relation between the direction of gradient at any point on the surface to the tangent plane at that point ?

- ☐ parallel
- ☒ perpendicular
- ☐ opposite direction
- ☐ No relation between them.

Question No: 10 (Marks: 1) - Please choose one

Two surfaces are said to intersect orthogonally if their norm common to them are

- ☒ perpendicular

Question No: 14 (Marks: 1) - Please choose one

The function $f(x, y) = \sqrt{y-x}$ is continuous in the region ----- and discontinuous elsewhere.

- ☐ $x \neq y$
- ☐ $x \leq y$
- ☒ $x > y$

$\therefore \sqrt{-1-2}$ $2-4 = -2$
 $\boxed{x > y}$ $4-2 = 2$

Question No: 15 (Marks: 1) - Please choose one

Plane is an example of -----

- ☐ Curve
- ☒ Surface
- ☐ Sphere
- ☐ Cone

Question No: 15 (Marks: 1) - Please choose one

Plane is an example of -----

- ☐ Curve
- ☒ Surface
- ☐ Sphere
- ☐ Cone

Question No: 16

Question # 4 of 30 (Start time: 10:38:02 AM, 26 June 2021)

Any vector whose magnitude or length is 1 is a _____ vector.

Select the correct option

☐	zero
☐	negative
☒	unit
☐	position

Question # 7 of 30 (Start time: 10:39:35 AM, 26 June 2021)

Position of any point in the plane can be obtained by the two perpendicular lines known as x and y axes and together we call it as _____ coordinates system.

Select the correct option

☐	Cartesian
☐	Spherical
☒	Rectangular
☐	Cylindrical



If $y = f(x)$ and $x = g(t)$ then by chain rule $\frac{dy}{dt} =$

Select the correct option

☐	$\frac{dy}{dx} - \frac{dx}{dt}$
☐	$\frac{dy}{dx} / \frac{dx}{dt}$
☐	$\frac{dy}{dx} + \frac{dx}{dt}$
☒	$\frac{dy}{dx} \cdot \frac{dx}{dt}$